

Efficient Hierarchical Skin Lesion Classification via Multi-Task Learning

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Abstract

The DERM12345 dataset contains 12,345 dermatoscopic images annotated with a three-tier taxonomy of 40 subclasses, 15 main classes, and 5 super classes. Conventional studies train separate models for each level, which increases computational cost and prevents shared representation learning. This study introduces a multi-task framework that attaches three classification heads to a single ResNet-34 backbone, enabling simultaneous prediction of subclass, main class, and super class categories. Compared with single-task baselines, the proposed multi-task model achieved competitive or superior performance while requiring only one model. These results demonstrate the effectiveness and efficiency of hierarchical multi-task learning for dermatoscopic image analysis.

Keywords: Dermatoscopy, Multi-task Learning, Skin Lesion Analysis, DERM12345 Dataset.

1. Introduction

Dermatoscopic datasets have contributed significantly to automated skin lesion classification, and DERM12345 provides one of the most comprehensive hierarchical taxonomies available, consisting of 40 subclasses, 15 main classes, and 5 super classes [1]. Previous works typically train independent models for each hierarchy level; However, this approach increases training time, requires additional storage, and fails to exploit the semantic relationships among the hierarchical labels.

To address these limitations, this work explores a multi-task learning strategy that uses a single backbone model to jointly classify all three levels. Because subclasses often share morphological characteristics with their corresponding main and super classes, jointly learning these labels is expected to improve generalization while reducing computational overhead.

2. Methods and Results

Experiments were conducted using the official DERM12345 dataset split, consisting of predefined training (80%) and test (20%) sets. From the training portion, 20% was further set aside as a validation set. Each image contains three labels simultaneously corresponding to the subclass, main class, and super class levels[1].

All images were resized to 224×224 pixels and normalized using standard ImageNet preprocessing. A ResNet-34 backbone pre-trained on ImageNet served as the shared feature extractor[2]. Three independent fully connected classification heads were attached to the final feature representation to predict subclass (40-way), main class (15-way), and super class (5-way) categories.

The model was trained for 50 epochs using the AdamW optimizer. Three learning rates (1×10^{-3} , 1×10^{-4} , 2×10^{-4}) were tested, and the model exhibiting the lowest validation loss was chosen for final evaluation. The overall loss function was defined as:

$$L_{total} = L_{subclass} + L_{main\ class} + L_{super\ class},$$

which enforces supervision across all hierarchy levels.

Figure 1 illustrates the proposed pipeline, where a single dermatoscopic image is encoded by ResNet-34, and all three hierarchical predictions are performed simultaneously predicted.

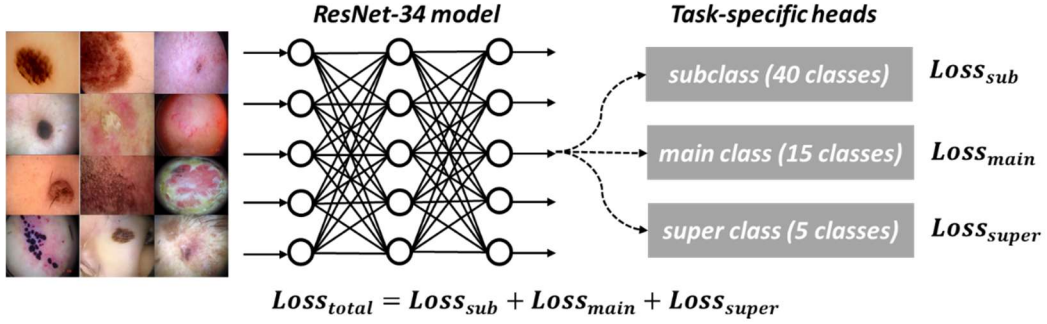


Figure 1. Illustration of the multi-task approach for hierarchical skin lesion classification.

Performance was compared with single-task baselines trained independently for each hierarchy level. The multi-task model achieved noticeable improvements in several metrics, particularly at the subclass level, which is the most challenging due to fine-grained differences among lesion types. The overall results in Table 1—including accuracy(Acc), AUPRC, sensitivity(Sen), precision(Pre), and F1-score(F1)—show that the multi-task framework either matched or surpassed single-task performance.

Table 1: Comparison of multi-level classification metrics for single-task and multi-task frameworks

	Subclass					Main class					Super class				
	Acc	AUPRC	Sen	Pre	F1	Acc	AUPRC	Sen	Pre	F1	Acc	AUPRC	Sen	Pre	F1
Single-task (subclass)	58.31	0.595	0.583	0.567	0.556	-	-	-	-	-	-	-	-	-	-
Single-task (main class)	-	-	-	-	-	63.06	0.656	0.631	0.620	0.622	-	-	-	-	-
Single-task (super class)	-	-	-	-	-	-	-	-	-	-	90.91	0.947	0.909	0.907	0.907
Multi-task	60.76	0.616	0.608	0.593	0.581	64.35	0.659	0.643	0.639	0.635	91.31	0.947	0.913	0.911	0.913

In addition, since only one model is required, the multi-task approach reduces storage requirements by approximately two-thirds and shortens training time. The improved subclass performance suggests that the model effectively captures hierarchical dependencies among the labels.

3. Conclusion

This study presents a multi-task approach for hierarchical skin lesion classification using the DERM12345 dataset. By training a single ResNet-34 backbone with three output heads, the method achieves competitive performance while significantly reducing computation and storage demands compared to training separate models. The gains at the subclass level emphasize the value of incorporating hierarchical supervision. Given its simplicity, efficiency, and strong performance, this approach offers a promising direction for real-world dermatology AI systems, particularly in resource-constrained clinical environments where training and storing multiple models may not be feasible.

4. References

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